TCAD-Based Investigation of GaN HEMTs for High-Efficiency Class-D Power Amplifier Applications

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***Abstract*— Gallium Nitride (GaN) high-electron-mobility transistors (HEMTs) have emerged as promising alternatives to conventional silicon MOSFETs for high-frequency and high-efficiency power electronic applications owing to their wide bandgap, high electron mobility, and low on-resistance. This paper presents a TCAD-based investigation of a GaN HEMT to evaluate its suitability for high-efficiency Class-D power amplifier applications. The electrical characteristics of the proposed device, including output and transfer characteristics, carrier concentration, and potential distribution, are analyzed to examine carrier transport, gate control, and the formation of a high-density two-dimensional electron gas (2DEG) at the AlGaN/GaN heterointerface. The TCAD results demonstrate reduced on-resistance and superior current-driving capability compared with conventional silicon (Si) MOSFET. Based on the extracted device parameters, an analytical loss model is employed to assess the potential impact of GaN technology on Class-D amplifier performance. The analysis indicates a significant reduction in conduction losses and projects an improvement in amplifier efficiency from 91.53% for the silicon-based implementation to nearly 97.0% for the GaN-based implementation under the adopted modeling assumptions. These results demonstrate the potential of GaN HEMTs for improving the efficiency of future Class-D power amplifiers while providing a device-level foundation for subsequent circuit-level simulation and experimental validation.**

Keywords- GaN HEMT, TCAD simulation, Class-D power amplifier, High-efficiency power electronics, Two-dimensional electron gas (2DEG).

I. INTRODUCTION

Power amplifiers (PAs) are indispensable components of modern electronic and communication systems, where they amplify low-power input signals to levels sufficient for driving loads such as antennas, loudspeakers, and other power-demanding devices. They play a vital role in wireless communication networks, radar and satellite systems, audio electronics, industrial equipment, medical instrumentation, and emerging Internet of Things (IoT) applications [1]. The performance of a power amplifier is primarily evaluated in terms of output power, power efficiency, linearity, and thermal reliability, as these parameters directly influence energy consumption, signal fidelity, and overall system performance. Among the various amplifier topologies, the Class-D power amplifier has gained considerable attention because of its switching-mode operation, which enables significantly higher efficiency than conventional linear amplifier classes while reducing power dissipation and heat generation [2, 3].

The increasing demand for high-frequency, high-power-density, and energy-efficient electronic systems has accelerated the adoption of wide-bandgap semiconductor technologies, particularly Gallium Nitride (GaN) and Silicon Carbide (SiC), for next-generation power electronic applications [4, 5]. Among these materials, GaN high-electron-mobility transistors (HEMTs) have emerged as promising candidates due to their wide bandgap, high electron mobility, high critical electric field, low on-resistance, and low parasitic capacitances, which collectively enable higher switching frequencies, reduced conduction and switching losses, and improved power conversion efficiency compared with conventional silicon devices [6-11]. Motivated by these advantages, this work presents a TCAD-based investigation of a GaN HEMT to evaluate its electrical characteristics and assess its suitability

for high-efficiency Class-D power amplifier applications. Device-level analyses, including output and transfer characteristics, carrier concentration, and potential distribution, are performed to investigate the underlying transport mechanisms, while an analytical loss model is employed to examine the potential efficiency improvement achievable over a conventional silicon-based implementation.

II. THEORETICAL MODELING

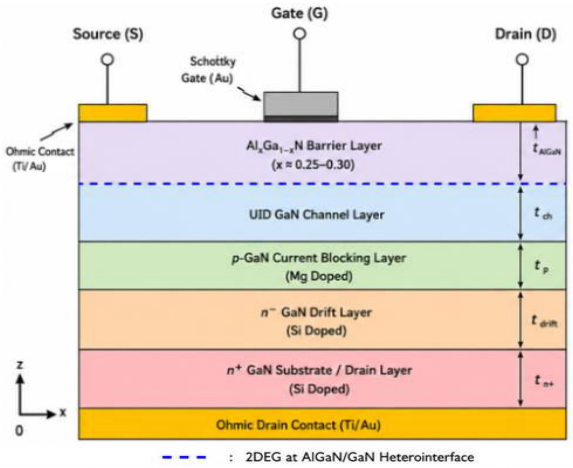


Figure 1. Schematic cross-sectional structure of the AlGaN/GaN HEMT used in TCAD simulations

Figure 1 shows the cross-sectional structure of the proposed AlGaN/GaN HEMT used in the TCAD simulations. The device consists of an AlGaN barrier, an undoped (UID) GaN channel, a Mg-doped p-GaN current-blocking layer, an Si-doped n^--GaN drift layer, and an n^+-GaN substrate with Ti/ Au-ohmic contacts and a Ni/Au Schottky gate. The corresponding material composition, doping concentrations, and layer thicknesses are summarized in Table 1. Polarization at the AlGaN/GaN heterointerface induces a high-density two-dimensional electron gas (2DEG), enabling high carrier mobility and low on-resistance for efficient high-frequency switching [6, 12, 13].

A. Class D Amplifier Operating Principle

A Class-D power amplifier achieves high power efficiency by operating its switching devices exclusively in the ON (saturation) and OFF (cut-off) states, thereby minimizing the simultaneous occurrence of high voltage and current across the switching device. Unlike linear amplifier classes, where significant power is dissipated during the active operating region, Class-D amplifiers employ pulse-width modulation (PWM) to convert the input analog signal into a high-frequency switching waveform. The amplified PWM signal is subsequently passed through a low-pass LC filter to reconstruct the amplified analog output while suppressing switching-frequency harmonics [2, 3].

The instantaneous duty cycle of the PWM signal is proportional to the input signal and is expressed as

$$D(t) = \frac{1}{2} + \frac{1}{2}\frac{V_{in}(t)}{V_{DD}} \quad (1)$$

where $D(t)$ is the instantaneous duty cycle, $V_{in}(t)$ represents the input signal, and V_{DD} represents the supply voltage. And the output filter cutoff frequency is determined by

$$f_c = \frac{1}{2\pi\sqrt{LC}} \quad (2)$$

where L and C are the inductance and capacitance of the output filter, respectively. Proper selection of the filter components ensures effective attenuation of switching harmonics while preserving the desired output signal.

B. Power Efficiency and Loss Modeling

Further, the efficiency of a Class-D power amplifier can be expressed as

$$\eta = \frac{P_{out}}{P_{in}} \times 100\% \quad (3)$$

where P_{in} and P_{out} are the input and output powers, respectively. Therefore, the total power loss of the amplifier can be expressed as

$$P_{loss} = P_{cond} + P_{sw} \quad (4)$$

where P_{cond} represents conduction loss and P_{sw} is the switching loss. And the P_{cond} associated with a power device can be expressed by

$$P_{cond} = I_{D,rms}^2 R_{Ds(on)} \quad (5)$$

where $I_{D,rms}^2$ is the r.m.s drain current and $R_{Ds(on)}$ is the on-state resistance of the switching device.

For the proposed AlGaN/GaN HEMT, the on-state resistance can be expressed as [6, 12]

$$R_{Ds(on)} = 2R_c + R_{acc,S} + R_{ch} + R_{acc,D} + R_{drift} \quad (6)$$

where R_c is the contact resistance, R_{ch} is the channel resistance, R_{acc} is the access resistance, and R_{drift} is the drift resistance.

Further,

$$R_{ch} = \frac{L_g}{q\mu_n n_s W} \quad (7)$$

where L_g is the effective channel length, q is the electron charge, μ_n is electron mobility in the 2DEG channel, n_s is the 2DEG sheet density, and W is the channel width.

Since GaN HEMTs typically show significantly lower on-state resistance than conventional Si-MOSFETs, a corresponding reduction in conduction loss is expected. In addition, the lower parasitic capacitances and absence of reverse-recovery charge in GaN HEMTs contribute to reduced switching losses during high-frequency operation [8, 9, 14, 15].

C. AlGaN/GaN HEMT Device Physics

The operation of the proposed HEMT device is based on the formation of a polarization-induced two-dimensional electron gas (2DEG) at the AlGaN/GaN heterointerface. Unlike conventional silicon MOSFETs, which require an inversion channel beneath the gate oxide, the AlGaN/GaN

HEMT utilizes spontaneous and piezoelectric polarization to generate a high-density electron channel without intentional channel doping [16-18]. The total polarization charge density can be expressed as

$$\sigma_{pol} = P_{sp} + P_{pz} \quad (8)$$

where P_{sp} and P_{pz} are the spontaneous and piezoelectric polarization components, respectively. Further, the corresponding 2DEG sheet carrier concentration can be expressed as

$$n_s = \frac{\sigma_{pol}}{q} \quad (9)$$

The low on-state resistance of the GaN HEMT is primarily attributed to the high electron mobility and large sheet carrier density of the 2DEG. The channel resistance is given by Equation (7).

D. TCAD-Based Modeling of the AlGaN/GaN HEMT

The proposed AlGaN/GaN HEMT was analyzed using TCAD simulation to investigate its electrical characteristics under steady-state operating conditions. The simulated device consists of an AlGaN barrier layer, an un doped (UID) GaN channel, a p-GaN current-blocking layer, and lightly and heavily doped GaN drift and drain regions. Material parameters, doping profiles, and polarization effects were incorporated to accurately model carrier transport within the heterostructure [6, 13, 15, 16].

The device characteristics were obtained by numerically solving Poisson's equation together with the carrier continuity equations under the drift-diffusion transport model. The electrostatic potential is governed by [19, 20]

$$\nabla(\varepsilon \nabla\phi) = -q(p - n + N_D^+ - N_D^-) \quad (10)$$

where ϕ represents the electrostatic potential, ε is the dielectric permittivity, n and p are the electron and hole concentrations, respectively. N_D^+ and N_D^- are the ionized donor and acceptor concentrations.

The electron drift–diffusion current equation can further expressed as

$$J_n = q\mu_n nE + qD_n\nabla_n \quad (11)$$

A similar expression is used for hole transport.

Poisson's Equation (10) was solved self-consistently with the electron and hole continuity equations under the drift–diffusion transport model (11) to obtain the output characteristics, transfer characteristics, carrier concentration, and electrostatic potential distributions. These simulated parameters were subsequently used to evaluate channel formation, carrier transport, gate control, and the on-state resistance of the proposed device for high-efficiency Class-D power amplifier applications.

In this work, the TCAD-extracted electrical characteristics of the GaN HEMT are combined with the above analytical framework to evaluate the potential efficiency improvement achievable in Class-D power amplifier applications. The quantitative performance assessment is presented and discussed in the subsequent Results and Discussion section.

Table 1. TCAD Simulation Parameters of the Proposed AlGaN/GaN HEMT [6, 7, 12, 17]

Parameter	***Values***
Al content (x)	0.25-0.30
Barrier thickness	25 nm
Channel thickness	2 um
p-GaN thickness	155 nm
Drift region thickness	5 µm
Mg doping (p-GaN)	1×10^{19}cm^{-3}
Si doping (n^-GaN)	5×10^{16}cm^{-3}
Si doping (n^+GaN)	1×10^{18}cm^{-3}

IV. RESULTS AND DISCUSSION

The TCAD-simulated output and transfer characteristics of the proposed vertical n-channel E-mode AlGaN/GaN HEMT are presented in Figures 2(a) and 2(b), respectively. Figure 2(a), shows that drain current (I_D) increases with the applied drain–source voltage (V_{DS}) for all gate bias conditions and further increases as the gate voltage (V_{GS}) becomes more positive. This behavior is characteristic of an enhancement-mode n-channel AlGaN/GaN HEMT, where a positive gate bias progressively restores the polarization-induced two-dimensional electron gas (2DEG) channel beneath the gate, thereby increasing the channel conductivity and enabling efficient carrier transport [6, 17]. Consequently, stronger carrier confinement within the triangular quantum well at the AlGaN/GaN heterointerface results in higher drain current. The high electron mobility and large sheet carrier concentration of the restored 2DEG significantly reduce the channel resistance, enabling efficient current conduction [6, 12]. The transfer characteristics shown in Figure 2(b) further demonstrate effective gate-controlled modulation of the drain current, confirming excellent electrostatic control of the channel and efficient switching operation. The monotonic increase in drain current with increasing gate voltage indicates high transconductance, which is highly desirable for high-speed switching and low-power operation. These characteristics demonstrate the suitability of the proposed vertical n-channel E-mode AlGaN/GaN HEMT for high-frequency, and high-efficiency Class-D power amplifier applications.

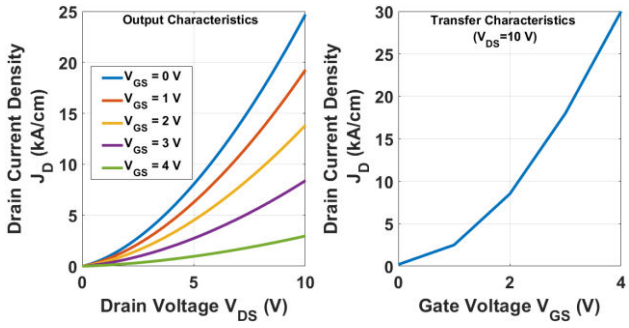


Figure 2. Simulated (a) output characteristics and (b) transfer characteristics of the proposed AlGaN/GaN HEMT obtained using TCAD.

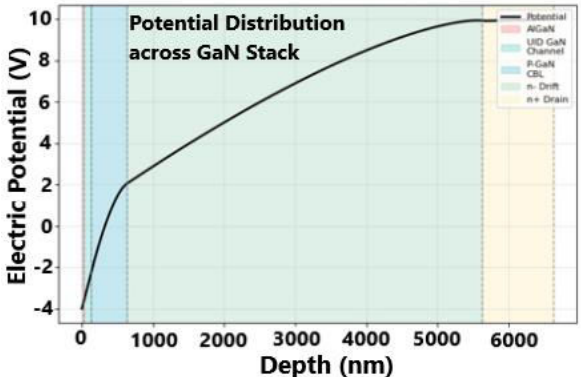


Figure 3. Electrostatic potential distribution of the proposed vertical n-channel E-mode AlGaN/GaN HEMT

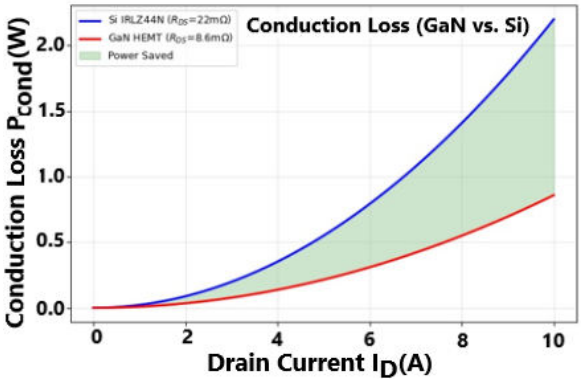


Figure 4. Conduction loss characteristics of the proposed vertical n-channel E-mode AlGaN/GaN HEMT

The electrostatic potential distribution and conduction loss characteristics of the proposed vertical n-channel E-mode AlGaN/GaN HEMT are presented in Figures 3 and 4, respectively. As shown in Figure 3, the electrostatic potential increases smoothly from +4 V at the gate contact to nearly +10 V at the drain contact, confirming proper implementation of the applied bias conditions and a well-distributed electric field across the AlGaN/GaN heterostructure. The gradual potential variation through the AlGaN barrier, UID-GaN channel, p-GaN current-blocking layer (CBL), and n^--GaN drift region demonstrates effective electrostatic control and uniform voltage sharing while minimizing localized electric-field crowding. Under the applied positive gate bias, the polarization-induced 2DEG channel is progressively restored beneath the gate, enabling efficient carrier transport through the device [12, 17]. The nearly linear potential variation across the lightly doped n^- drift region further indicates efficient voltage blocking capability, making the proposed device suitable for high-voltage and high-frequency power-switching applications [16, 17].

The impact of this device structure on power dissipation is shown in Figure 4, which compares the conduction losses of the proposed GaN HEMT with those of a conventional Si-MOSFET (IRLZ44N). Due to its significantly lower on-state resistance ($R_{Ds(on)}$= 8.6mΩ) compared with the silicon counterpart (22 mΩ), the GaN HEMT shows substantially lower conduction losses over the entire operating current range. At a drain current of 10 A, the conduction loss is reduced from 2.2 W for the Si MOSFET to about 0.86 W for the GaN HEMT, corresponding to a reduction of nearly 61%. This considerable reduction in power dissipation results directly from the high-mobility 2DEG channel and lower channel resistance of the AlGaN/GaN HEMT, leading to improved efficiency, reduced thermal stress, and enhanced power density. The combined electrostatic and conduction-loss characteristics therefore demonstrate the suitability of the proposed vertical n-channel E-mode AlGaN/GaN HEMT for high-frequency, and high-efficiency Class-D power amplifier applications.

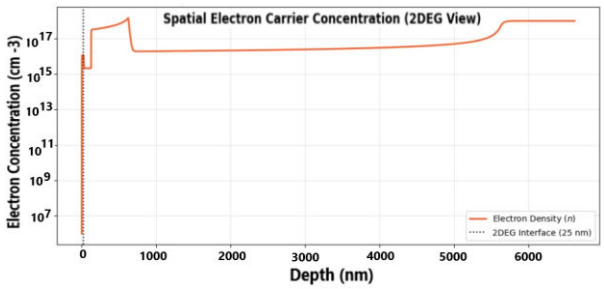


Figure 5. TCAD-simulated spatial electron concentration profile of the proposed vertical n-channel E-mode AlGaN/GaN HEMT

Figure 5 presents the TCAD-simulated spatial electron concentration profile of the proposed vertical n-channel E-mode AlGaN/GaN HEMT. A pronounced electron concentration peak at the AlGaN/GaN heterointerface confirms the formation of the polarization-induced 2DEG electron gas. Under positive gate bias, the 2DEG is progressively restored at the heterointerface, resulting in strong carrier confinement within the triangular quantum well and efficient electron transport [12]. The restored 2DEG shows high electron mobility and sheet carrier concentration, thereby reducing the channel resistance and enhancing the current-driving capability. Away from the heterointerface, the electron concentration decreases according to the device doping profile. These results confirm effective 2DEG restoration and demonstrate the suitability of the proposed device for low-loss, high-efficiency Class-D power amplifier applications.

Table 2. Comparison of Conventional Si MOSFET and Proposed AlGaN/GaN HEMT [4, 5, 7-9, 16]

Parameter	***Si-MOSFET (IRLZ44N)***	***Proposed AlGaN/GaN HEMT***
Channel mechanism	Inversion channel	Gate-controlled polarization-induced 2DEG
Bandgap	1.12eV	3.4eV
Breakdown voltage	55V	> 600 V
$R_{Ds(on)}$	22mΩ	8.6mΩ
Threshold voltage	1-2 V	+3.6 V
switching frequency	Moderate	Very high

Table 2 summarizes the performance comparison between the conventional Si MOSFET (IRLZ44N) and the proposed AlGaN/GaN HEMT. The proposed device achieves a substantially lower on-state resistance of 8.6 mΩ compared with 22 mΩ for the Si MOSFET, resulting in reduced conduction losses and improved power efficiency. In addition, the wide bandgap of GaN enables a breakdown voltage exceeding 600 V, which is significantly higher than the 55 V breakdown voltage of the silicon counterpart. The proposed HEMT also shows superior transconductance and supports much higher switching frequencies, making it well suited for high-frequency, high-efficiency Class-D power amplifier applications. Overall, these results demonstrate the clear performance advantages of GaN technology over conventional silicon devices for next-generation power electronic systems [4, 5, 7-9, 16].

To further assess the effectiveness of the proposed device, its key electrical characteristics are benchmarked against representative AlGaN/GaN HEMTs reported in the literature. The comparison focuses on critical performance parameters, including breakdown voltage, on-state resistance, operating mode, and intended application, thereby providing a broader perspective on the proposed device relative to existing GaN technologies.

Table 3. Benchmark comparison of the proposed AlGaN/GaN HEMT with representative AlGaN/GaN HEMTs reported in the literature.

Reference	***Device Structure***	***VBD (V)***	***RDS (on)***	***Operation***	***Application***
Wu et al., 2001[22]	AlGaN/GaN, HEMT	200	20 mΩ	D-mode	RF
Chowdhury &Mishra, 2013[17]	Vertical GaNHEMT	600	12 mΩ	Vertical	Power
Amano et al., 2018[7]	GaN Power Device	600	10-15 mΩ	Power	Power Electronics
This Work	Vertical E-modeAlGaN/GaN HEMT	600	8.6 mΩ	E-mode	Class-D PA

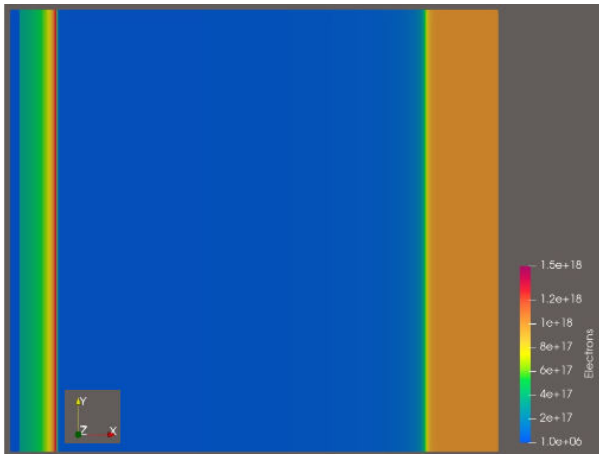


Figure 6. TCAD-simulated electron concentration contour of the proposed vertical n-channel E-mode AlGaN/GaN HEMT

Figure 6 presents the simulated electron concentration contour of the proposed vertical n-channel E-mode AlGaN/GaN HEMT obtained from TCAD. A pronounced peak in electron concentration is observed at the AlGaN/GaN heterointerface, confirming the formation of the polarization-induced two-dimensional electron gas (2DEG). The electron density at this interface exceeds 10^{18} cm^{-3}, indicating strong carrier confinement within the quantum well and the formation of a highly conductive channel responsible for efficient electron transport.

The AlGaN barrier demonstrates a very low free-electron concentration because of its undoped nature, while the adjacent p-GaN current-blocking layer remains electron depleted owing to Mg acceptor doping [17]. Away from the heterointerface, the electron concentration decreases substantially throughout the lightly doped n^--GaN drift region, which is intentionally designed to sustain high blocking voltage while minimizing electric-field crowding. However, the heavily doped n^+-GaN drain region maintains a uniformly high electron concentration of about 10^{18} cm^{-3}, ensuring the formation of a low-resistance ohmic contact. The simulated carrier distribution therefore confirms proper 2DEG formation, effective carrier confinement, and the intended doping profile, all of which contribute to the low on-state resistance, high current-driving capability, and superior switching performance of the proposed AlGaN/GaN HEMT.

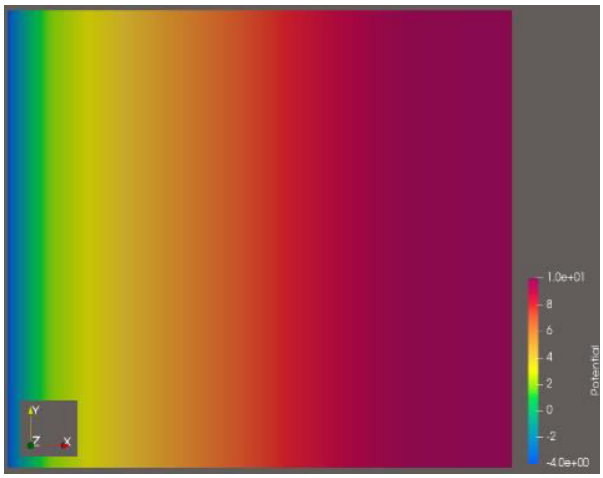


Figure 7. TCAD -simulated electrostatic potential distribution of the proposed vertical n-channel E-mode AlGaN/GaN HEMT

Figure 7 demonstrates the TCAD-simulated electrostatic potential distribution across the proposed vertical n-channel E-mode AlGaN/GaN HEMT under the applied bias conditions. The electrostatic potential varies smoothly from about +4 V at the gate contact to nearly +10 V at the drain contact, confirming proper implementation of the applied boundary conditions. The gradual increase in potential through the AlGaN barrier, UID-GaN channel, p-GaN current-blocking layer (CBL), and n^--GaN drift region

indicates effective electrostatic control and uniform voltage sharing throughout the device. The nearly linear potential variation across the lightly doped n^- drift region demonstrates that the applied drain voltage is distributed over the entire drift layer, thereby minimizing localized electric-field crowding and enhancing the breakdown capability of the device [16, 17, 21]. Finally, the constant potential within the heavily doped n^+-GaN drain region confirms the formation of a low-resistance ohmic contact. The simulated potential profile therefore validates the proposed device design and demonstrates its suitability for high-voltage, high-frequency, and high-efficiency power-switching applications.

V. CONCLUSION

This paper presents a TCAD-based investigation of a proposed vertical n-channel E-mode AlGaN/GaN high-electron-mobility transistor (HEMT) for high-efficiency Class-D power amplifier applications. The simulated output and transfer characteristics confirmed effective gate control, high transconductance, and efficient current transport through the polarization-induced two-dimensional electron gas (2DEG) formed at the AlGaN/GaN heterointerface. The simulated electron concentration and electrostatic potential distributions further verified proper 2DEG formation, carrier confinement, uniform voltage sharing, and efficient electrostatic control of the device. Due to its low on-state resistance (8.6 mΩ), the proposed GaN HEMT achieved about 61% lower conduction loss than the conventional Si-MOSFET (IRLZ44N), resulting in a projected Class-D amplifier efficiency improvement from 91.53% to nearly 97.0% under the adopted analytical model. These results demonstrate the potential of the proposed AlGaN/GaN HEMT for high-frequency, high-efficiency power amplifier applications. Future work will focus on circuit-level implementation, experimental validation, and switching-loss analysis under practical operating conditions.

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